THE CURE CENTER
FOR ULTRASOUND RESEARCH & EDUCATION

# ABDOMINAL AORTRA & IVC/ GREATER VESSELS ULTRASOUND SYLLABUS

**COURSE CODE: DMS 200**

**ABDOMINAL AORTA AND IVC/GREATER VESSELS ULTRASOUND**

**Instructor:**
**Telephone:**
**E-Mail:**
**Office Hours:**
**Mon, Wed, Fri:** 12pm to 1pm & 5pm to 6pm
Tues: 9am – 5pm
Thurs: 11am – 5pm

**Start Date:**

**Completion Date:**

**Days and Times:**
Monday, Wednesday and Friday

**Course Hours:**
**Morning Session:**
8:00 am -10:00am &
10:10am – 12:00 pm
**Afternoon Sessions:**
1:00pm – 3:00pm &
3:10 pm – 5:00 pm

**Class Location:**
Classroom #:
Lab#:

**Prerequisites:**
BIO 100 Introduction to A&P
PAT 100 Patient Care
LAW 100 Medical Law & Ethics

**Course Contact Hours: 40**
**Course Length : 5days**
**Lecture: 20 hours**
**Laboratory: 20 hours**
**Externship:**
**Course Delivery: Residential**
**Pre-requisite: None**

**Americans with Disabilities Act Guidelines:**

Requests for accommodations should be submitted to the campus president. Faculty are not authorized to grant accommodations.

## COURSE DESCRIPTION

This is a course directed towards developing the principles, techniques, fundamental knowledge and skills of each student necessary to function as a qualified diagnostic medical sonographer in clinical and diagnostic practice of abdomen sonography.
In this course lectures, projects, discussions and practical visual aids using The CURE campus Lab equipment, will be used to identify normal and abnormal abdominal anatomy and physiology.

Students will learn patient positioning, and different abdominal ultrasound protocols.
Laboratory results, and the role of different complementary imaging modalities will be discussed dependence on pertinent different abdominal diseases.

Special structures such as thyroid, parathyroid, scrotum, prostate, breast, and superficial masses will be investigated in further detail to include biopsy and aspiration techniques as well as the use of Pulsed Doppler and Color Doppler evaluation. Ocular, musculoskeletal and lung sonography will be discussed.

Emphasis will be placed on the academic and laboratory applications within the highly specialized area of general abdomen sonography including basic patient care principles and professionalism.

## Instructional Materials

**1. Textbook**

Title: Textbook of Diagnostic Sonography, 8th Edition (2 Volume Set)
Author(s): Sandra L. Hagen-Ansert, MS, RDMS, RDCS, FASE, FSDMS
Publisher: Mosby; 8th edition (May 3, 2017)
ISBN: 978-0323353755

**2. Workbook**
Title: Workbook for Diagnostic Sonography, 8th Edition
Author(s): Sandra L. Hagen-Ansert, MS, RDMS, RDCS, FASE, FSDMS
Publisher: Mosby; 8th edition (May 3, 2017)
ISBN: 978-0323441834

3. **Reference Book**
Title: Pocket Protocols for Sonography Scanning, 4th Edition
Author(s): Betty Bates-Tempkin, BA, RT®, RDMS
Publisher: Saunders; 4th edition (December 27, 2014)
ISBN: 978-1455773220

## Equipment/Technology

- LCD Projector
- Laptop Computer
- Mindray Ultrasound Machine

## Material

- Electronic and Printed Worksheets
- Textbooks
- Visual Aids: DVD Videos, Case Study Reviews, Demonstrations and Hands on Clinical Laboratory Practice.

During clinical laboratory sessions various techniques such as role playing, and scanning exercises will be used.

# ABDOMINAL AORTRA & IVC/ GREATER VESSELS ULTRASOUND SYLLABUS

**Grade Point Value**

| Grade | Percentile Range | Points |
|---|---|---|
| A+ | 97-100 | 4.00 |
| A | 93-96.99 | 3.85 |
| A- | 90. -92.99 | 3.67 |
| B+ | 87-89.99 | 3.33 |
| B | 83-86.99 | 3.00 |
| B- | 80-82.99 | 2.67 |
| C+ | 77-79.99 | 2.33 |
| C | 73-76.99 | 2.00 |
| C- | 70-72.99 | 1.67 |
| D+ | 67-69.99 | 1.33 |
| D | 63-66.99 | 1.00 |
| D- | 60-62.99 | .67 |
| F | Below 60-00 | 0 |

**Methods of Evaluation**

| | |
|---|---|
| Exams / Quizzes/ Midterm/ Final | 40% |
| Projects/ Attendance Participation/ Homework Assignments | 20% |
| Scan-out | 40% |
| | 100% |

**Teaching Strategies**

A variety of teaching strategies may be utilized in this course, including but not limited to: lecture, discussion, written classroom exercises, visual aids, videos, case study reviews, group and/or individual projects, quizzes, exams, reading assignments, writing assignments, interactive exercises demonstrations and hands, internet research and hands on clinical laboratory practice.

**Examination Standards**

Students are expected to be honest in all academic work. Cheating of any kind is not tolerated. Violations of this principle will result in a grade of zero in the course and additional disciplinary action including expulsion as determined by CURE

## Course Objectives:

Upon successful completion of this course, the student should be able to:

- Integrate theory into sonographic practice and sharpen critical thinking and affective skills necessary to function as a qualified diagnostic medical sonographer.
- Describe the anatomy of the arterial system, and venous system.
- Understand the function of the circulatory system.
- Recognize the sonographic findings and pathology found in the vascular structures.
- Define the two types of aneurysm formation.
- Explain the factors that may cause development of an aneurysm.
- Describe the anatomy of the portal venous system.
- Identify the sonographic findings in aortic dissection.
- Identify causes of pseudo pulsatile abdominal masses.
- Explain the sonographic findings and complications in portal hypertension.
- Describe the Doppler flow patterns in arterial versus venous structures.
- Describe normal anatomy of the, including vascular supply and relational landmarks.
- Perform sonographic evaluation of the aorta and the IVC.
- Successfully perform the Aorta and IVC protocol, independently.
- pass the didactic course with a minimum passing grade of 73.

# ABDOMINAL AORTRA & IVC/ GREATER VESSELS ULTRASOUND SYLLABUS

## Attendance Policy

Routine, regular class attendance is a requirement for this course.
Promptness and dependability are the soft skills that garner special attention

## Attendance Rubric:

Up to 25% will be deducted for each hour missed or for any part thereof (5% for each 10 minutes missed).
Students are required to meet their professor to discuss matters of tardiness or absence.

Students are required to notify their instructor via E mail or voicemail to communicate an absence or lateness from class

A student absent from all of his or her classes for a period of 14 consecutive calendar days during which classes or examinations are in session will be regarded as having withdrawn from the program and will receive a grade of W for each course in which he or she is enrolled.

## HOMEWORK

An essential part of every program of study, homework enables students to master course learning objectives and leads toward the achievement of the overall program objectives. Homework refers to self-directed learning activities such as reading assignments, research activities and projects, quiz/exam preparation, knowledge checks and self-assessments, etc.

## Late Homework/Make-Up Policy

Late homework will be accepted under the following guidelines:

1. It must be turned in within two business days (Monday through Thursday) after the due date.
2. However, homework turned in after the due date will be penalized 10% for each day it is late up to two calendar days after the due date.
3. Homework submitted after three business days after the due date will earn a zero.

**Exams and quizzes** - Make up of a quiz or an exam will be allowed under the following extenuating circumstances only:

- Documented illness
- Military duty
- Death of a family member
- Court appearance or jury duty

Documentation of the above circumstances must be provided to the Program Director. ***The ability to make-up a quiz or exam must be approved by the Program Director.***

**All assignments should be submitted directly to the instructor.**

## Classroom Policy

Our goal is to build a respectful learning and work environment that allows for positive communication and teamwork. To support your development, all students must abide by academic policies related to attendance, behavior and professional conduct as published in the current institutional catalog. Special attention is granted to the following items:

- There is to be no eating or drinking in any lab/computer class.
- There is to be no eating in ANY class.

- Cellular phones may only be used during class if/when advised by the instructor and required for a learning activity.
- All students must be in full uniform which includes white or black shoes at **ALL TIMES.**
- **There are no rescheduled exams *unless specified by the professor* with advance notification.**
- The determination to accept late homework assignments is made only by the instructor and occurs only in the event of extraordinary circumstances. If the professor does allow for an extension on assignments, there will be an automatic daily reduction by 10 points, each day beyond the original due date
- All classes are expected to start and end at a predetermined start and end time.

**Accommodations for Students with Disabilities**

If you have a special learning need, please contact the Program Director. The Program Director and the Campus President will work with the student to identify reasonable accommodations or adjustments necessary to enable him or her to fully participate in the educational processes.

**Professional Responsibility**

Students are expected to aspire to the same high level of ethics and professionalism that is demanded of registered Ultrasound students. This requires students to conduct themselves as professionals and to fully engage in the learning process by preparing for and participating in all classroom, laboratory and clinical activities.

Attendance is taken each hour of class by the instructor and turned over to the Registrar before the end of the class day. Early departures, tardiest, and class cuts will be recorded in quarter-hour increments and a proportional grade reduction will also be applied. A period of less than 15 minutes will be counted as a quarter-hour of absence. Attendance records are maintained by the Registrar as part of the student's permanent academic record.

# ABDOMINAL AORTRA & IVC/ GREATER VESSELS ULTRASOUND SYLLABUS

**COURSE OUTLINE**
Below is the course schedule which identifies lecture topics, assignments and other course activities and the homework required for completion each week.

The course syllabus is a general plan for the course; deviations may be necessary and will be announced.

| Lecture Topic | In-Class Activity/Lab Assignment | Homework/Out of Class |
|---|---|---|
| **Week One Objectives:**<br>Introduction to course, class policies, and Syllabus.<br>• Describe the anatomy of the arterial system, and venous system.<br>• Understand the function of the circulatory system.<br>• Recognize the sonographic findings and pathology found in the vascular structures.<br>• Define the two types of aneurysm formation.<br>• Explain the factors that may cause development of an aneurysm. | | |
| **DAY 1.**<br>**Function of the Circulatory System**<br>Aorta<br>Root of the Aorta<br>Ascending Aorta<br>Descending Aorta<br>Abdominal Aorta | Power Point Presentation/ Lecture | **Read**<br>Textbook: Diagnostic Sonography 8ed<br>Sandra Hagen<br>**Ultrasound Scanning, Principles and Protocols 4th Edition,**<br>Betty Bates-Tempkin<br><br>Chapter 8: Vascular System<br><br>Pages. 141-189<br>Chapter 8 Homework:<br>Workbook Assignments. Exercise 1 and Exercise 2<br>Page 43 |
| **DAY 2:**<br><br>Aorta and Iliac Artery Protocol<br>Common Iliac Arteries<br>Abdominal Aortic Branches<br>Pathology of the Aorta | Power Point Presentation/ Lecture | **Read**<br>Textbook: Diagnostic Sonography 8ed<br>Sandra Hagen<br>**Ultrasound Scanning, Principles and Protocols 4th Edition,**<br>Betty Bates-Tempkin<br><br>Chapter 8: Vascular System<br><br>Pages. 141-189<br>Chapter 8 Homework:<br>Workbook Assignments. Exercise 3 and Exercise 4<br>Page 44 |
| **DAY 3.**<br>**Inferior Vena Cava**<br>Inferior Vena Cava Abnormalities<br>Lateral Tributaries to the Inferior Vena Cava<br>Anterior Tributaries to the Inferior Vena Cava | Power Point Presentation/ Lecture | **Read**<br>Textbook: Diagnostic Sonography 8ed<br>Sandra Hagen<br>**Ultrasound Scanning, Principles and Protocols 4th Edition,**<br>Betty Bates-Tempkin<br><br>Chapter 8: Vascular System<br><br>Pages. 141-189<br>Chapter 8 Homework: |

| | | |
|---|---|---|
| | | Workbook Assignments. Exercise 5 & Exercise 6<br>Page 45-47 |

| Lecture Topic | In-Class Activity/Lab Assignment | Homework/Out of Class |
|---|---|---|
| **Week TWO Objectives:**<br>• Describe the anatomy of the portal venous system.<br>• Identify the sonographic findings in aortic dissection.<br>• Identify causes of pseudo pulsatile abdominal masses.<br>• Explain the sonographic findings and complications in portal hypertension.<br>• Describe the Doppler flow patterns in arterial versus venous structures.<br>• Describe normal anatomy of the, including vascular supply and relational landmarks.<br>• Perform sonographic evaluation of the aorta and the IVC.<br>• Successfully perform the Aorta and IVC protocol, independently. | | |
| **DAY 4.**<br>**Portal Venous System**<br>Portal Vein<br>Splenic Vein<br>Superior Mesenteric Vein<br>Inferior Mesenteric Vein | Power Point Presentation/ Lecture | **Read**<br>Textbook: Diagnostic Sonography 8ed<br>Sandra Hagen<br>**Ultrasound Scanning, Principles and Protocols 4th Edition,**<br>Betty Bates-Tempkin<br><br>Chapter 8: Vascular System<br><br>Pages. 141-189<br>Chapter 8 Homework:<br>Workbook Assignments. Exercise 7 and Exercise 8<br>Pages 47-48 |
| **DAY 5:**<br><br>**Abdominal Doppler Techniques**<br>Blood Flow Analysis<br>Doppler Technique<br>Doppler Flow Patterns in the Abdominal Arterial Vessels<br>Doppler Flow Patterns in the Abdominal Venous Vessels<br>Doppler Flow Patterns in the Abdominal Venous Vessels<br><br>REVIEW/SCANOUT/FINAL | Power Point Presentation/ Lecture | **Read**<br>Textbook: Diagnostic Sonography 8ed<br>Sandra Hagen<br>**Ultrasound Scanning, Principles and Protocols 4th Edition,**<br>Betty Bates-Tempkin<br><br>Chapter 8: Vascular System<br>Pages. 141-189<br>Chapter 8 Homework:<br>Workbook Assignments. Exercise 9, Exercise 10 and Exercise 11<br>Pages 49-52<br><br>FINALEXAM/ SCANOUT |